



Department of Civil Engineering

Academic Year: 2021 - 2022 (Odd Semester)

Degree, Semester & Branch: III semester B.E Civil Engineering

Course Code & Title: CE8392 & Engineering Geology

Name of the Faculty member: Mr.J.Ramprashath

Innovative Practice Description

Unit / Topic: Unit - III / Classification of Rocks

- **Course Outcome:** CO3
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** Cross Word Puzzle
- **Justification:**

The wide variety of rocks found on and below the surface of the earth are majorly classified into three types of rocks namely, igneous, sedimentary and metamorphic based on their formation. Hence, I made a cross word puzzle on the above classification of rocks to make the students more familiar with the various types of rocks, their mode of occurrence, engineering properties, etc.

- **Time Allotted for the Activity:** 10 minutes.
- **Details of the Implementation:**
 - i. Cross Word Puzzle is a very commonly used classroom assessment technique. It is a fun filled assessment technique which also engage the students actively in the class.
 - ii. I took all the students to Civil Cadd lab and asked the students to take the cross word puzzle.

- **CO – PO / PSO mapping:**

CO	PO1	PO4	PO5	PO6	PO12	PSO1
CO1	2	1	1	1	1	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO4
Justification for correlation	By understanding the formation of rocks, its types and its engineering properties, the students will be able to provide solutions for complex civil engineering problems by applying the geological knowledge.	Research studies on petrology helps in formulating suitable conclusion for investigations of complex civil engineering problems.

	PO5	PO6
	Various modern tools are used in determining the engineering properties of rocks in conducting laboratory as well as field experiments.	Geological knowledge on rocks is very much essential for site selection and design of various civil engineering structures like dams, tunnels etc. which will in turn ensure public safety.
	PO12	PSO1
	As the field of petrology and rock mechanics are vast, lifelong learning is necessary to explore the stream further.	Petrology and rock mechanics knowledge are important criteria for presenting the civil engineering projects effectively.

• **Images / Screenshot of the practice:**



Crossword Labs

[Make a Crossword](#) [Find a Crossword](#) [About](#) [Login/Sign Up](#)

Classification of Rocks



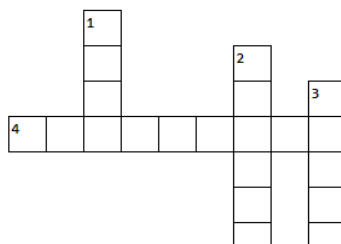
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Across

- 4. Volcanic Rocks
- 6. Plutonic Rocks
- 9. Secondary Rocks
- 10. Second Direction of easiest parting

Down

- 1. Direction of easiest parting
- 2. Rocks G = 2.6
- 3. Heat and Pressure
- 5. rocks G = 3.6
- 6. Rocks G = 2.8 TO 2.9
- 7. Volcanic Origin

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

The students felt easy to remember and recollect the concepts involved in the given topic which will make them to attend the questions under this topic in Internal Assessment Test.

- ❖ ***Benefit of the practice:*** (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

- i. Cross Word Puzzle is an effective way of involving all students in class simultaneously.
- ii. It prompts students to reflect on the day's lesson and provides the instructor with useful feedback.

- ❖ ***Challenges faced in implementation:***

Initially, I have planned the activity for 10 minutes but students have taken 15 to 20 minutes to complete.

References:

- ❖ Parbin Singh. A "Text book of Engineering and General Geology", Katson publishing house, Ludhiana 2009.
- ❖ <https://crosswordlabs.com/view/classification-of-rocks>